

Benchmarking Results (terminal screenshots)

Machine A - Laptop

- **CPU1:** 13th Gen Intel(R) Core(TM) i5-13420H
 - **Cores:** 8 cores, 12 logical processors
- **GPU1:** NVIDIA GeForce RTX 4050
 - **VRAM:** 6GB
 - **CUDA version:** 12.8

Machine B - Workstation

- **CPU2:** Intel(R) Xeon(R) w5-3435X
 - **Cores:** 16 cores, 32 logical processors
- **GPU2:** NVIDIA RTX 4000 Ada Generation
 - **VRAM:** 20GB
 - **CUDA version:** 12.8

FP32 (CPU1)

640 input size

```
(yolo_env) PS C:\Users\rjrao\Desktop\P_Det> python code_files/main_execution.py --model models/yolo_onnx_models/F
P32_onnx_models/640/yolov8n_640.onnx --device cpu --input-size 640 --data-root imagedata
FPS: 23.9696
Model size (on disk): 12.26 MiB (12,851,087 bytes)
mAP@IoU=0.50 : 0.681449
(yolo_env) PS C:\Users\rjrao\Desktop\P_Det>

mAP@IoU=0.50 : 0.681449
(yolo_env) PS C:\Users\rjrao\Desktop\P_Det> python code_files/main_execution.py --model models/yolo_onnx_models/F
P32_onnx_models/640/yolov8s_640.onnx --device cpu --input-size 640 --data-root imagedata
FPS: 10.3011
Model size (on disk): 42.79 MiB (44,869,881 bytes)
mAP@IoU=0.50 : 0.709116
(yolo_env) PS C:\Users\rjrao\Desktop\P_Det>

(yolo_env) PS C:\Users\rjrao\Desktop\P_Det> python code_files/main_execution.py --model models/yolo_onnx_models/F
P32_onnx_models/640/yolov8m_640.onnx --device cpu --input-size 640 --data-root imagedata
FPS: 1.9216
Model size (on disk): 99.00 MiB (103,809,586 bytes)
mAP@IoU=0.50 : 0.723569
(yolo_env) PS C:\Users\rjrao\Desktop\P_Det>
```

320 input size

```
(yolo_env) PS C:\Users\rjr\ao\Desktop\P_Det> python code_files/main_execution.py --model models/yolo_onnx_models/F
P32_onnx_models/320/yolov8n_320.onnx --device cpu --input-size 320 --data-root imagedata
FPS: 64.1732
Model size (on disk): 12.14 MiB (12,725,005 bytes)
mAP@IoU=0.50 : 0.521438
(yolo_env) PS C:\Users\rjr\ao\Desktop\P_Det>
dels\640\yolov8s_320.onnx
(yolo_env) PS C:\Users\rjr\ao\Desktop\P_Det> python code_files/main_execution.py --model models/yolo_onnx_models/F
P32_onnx_models/320/yolov8s_320.onnx --device cpu --input-size 320 --data-root imagedata
FPS: 35.7109
Model size (on disk): 42.67 MiB (44,743,799 bytes)
mAP@IoU=0.50 : 0.614547
(yolo_env) PS C:\Users\rjr\ao\Desktop\P_Det>
(yolo_env) PS C:\Users\rjr\ao\Desktop\P_Det> python code_files/main_execution.py --model models/yolo_onnx_models/F
P32_onnx_models/320/yolov8m_320.onnx --device cpu --input-size 320 --data-root imagedata
FPS: 2.8104
Model size (on disk): 98.88 MiB (103,683,476 bytes)
mAP@IoU=0.50 : 0.660944
(yolo_env) PS C:\Users\rjr\ao\Desktop\P_Det>
```

1280 input size

```
(yolo_env) PS C:\Users\rjr\ao\Desktop\P_Det> python code_files/main_execution.py --model models/yolo_onnx_models/F
P32_onnx_models/1280/yolov8n_1280.onnx --device cpu --input-size 1280 --data-root imagedata
FPS: 5.9072
Model size (on disk): 12.74 MiB (13,355,362 bytes)
mAP@IoU=0.50 : 0.725778
(yolo_env) PS C:\Users\rjr\ao\Desktop\P_Det>
(yolo_env) PS C:\Users\rjr\ao\Desktop\P_Det> python code_files/main_execution.py --model models/yolo_onnx_models/F
P32_onnx_models/1280/yolov8s_1280.onnx --device cpu --input-size 1280 --data-root imagedata
FPS: 2.5601
Model size (on disk): 43.27 MiB (45,374,156 bytes)
mAP@IoU=0.50 : 0.761152
(yolo_env) PS C:\Users\rjr\ao\Desktop\P_Det>
(yolo_env) PS C:\Users\rjr\ao\Desktop\P_Det> python code_files/main_execution.py --model models/yolo_onnx_models/F
P32_onnx_models/1280/yolov8m_1280.onnx --device cpu --input-size 1280 --data-root imagedata
FPS: 0.9990
Model size (on disk): 99.48 MiB (104,313,941 bytes)
mAP@IoU=0.50 : 0.757325
(yolo_env) PS C:\Users\rjr\ao\Desktop\P_Det>
```

FP16 quantized (CPU1)

640 input size

```
(yolo_env) PS C:\Users\rjr\ao\Desktop\P_Det> python code_files/main_execution.py --model models/yolo_onnx_models/F
P16_quantized_models/640/yolov8n_640_fp16.onnx --device cpu --input-size 640 --data-root imagedata
C:\Users\rjr\ao\Desktop\P_Det\code_files\postprocessing.py:34: RuntimeWarning: overflow encountered in scalar mult
iply
    inter_area = (x2 - x1).clip(min=0) * (y2 - y1).clip(min=0)
C:\Users\rjr\ao\Desktop\P_Det\code_files\postprocessing.py:66: RuntimeWarning: invalid value encountered in scalar
divide
    return inter_area / union_area
FPS: 13.5533
Model size (on disk): 6.14 MiB (6,436,157 bytes)
mAP@IoU=0.50 : 0.681476
(yolo_env) PS C:\Users\rjr\ao\Desktop\P_Det>
```

```

(yolo_env) PS C:\Users\rjrao\Desktop\P_Det> python code_files/main_execution.py --model models/yolo_onnx_models/F
P16_quantized_models/640/yolov8s_640_fp16.onnx --device cpu --input-size 640 --data-root imagedata
C:\Users\rjrao\Desktop\P_Det\code_files\postprocessing.py:34: RuntimeWarning: overflow encountered in scalar mult
iply
    inter_area = (x2 - x1).clip(min=0) * (y2 - y1).clip(min=0)
C:\Users\rjrao\Desktop\P_Det\code_files\postprocessing.py:66: RuntimeWarning: invalid value encountered in scalar
divide
    return inter_area / union_area
FPS: 6.7362
Model size (on disk): 21.41 MiB (22,445,539 bytes)
mAP@IoU=0.50 : 0.709287

(yolo_env) PS C:\Users\rjrao\Desktop\P_Det> python code_files/main_execution.py --model models/yolo_onnx_models/F
P16_quantized_models/640/yolov8m_640_fp16.onnx --device cpu --input-size 640 --data-root imagedata
C:\Users\rjrao\Desktop\P_Det\code_files\postprocessing.py:34: RuntimeWarning: overflow encountered in scalar mult
iply
    inter_area = (x2 - x1).clip(min=0) * (y2 - y1).clip(min=0)
C:\Users\rjrao\Desktop\P_Det\code_files\postprocessing.py:66: RuntimeWarning: invalid value encountered in scalar
divide
    return inter_area / union_area
FPS: 3.2922
Model size (on disk): 49.51 MiB (51,917,808 bytes)
mAP@IoU=0.50 : 0.723501
(yolo_env) PS C:\Users\rjrao\Desktop\P_Det>

```

320 input size

```

(yolo_env) PS C:\Users\rjrao\Desktop\P_Det> python code_files/main_execution.py --model models/yolo_onnx_models/F
P16_quantized_models/320/yolov8n_320_fp16.onnx --device cpu --input-size 320 --data-root imagedata
FPS: 61.7969
Model size (on disk): 6.08 MiB (6,373,145 bytes)
mAP@IoU=0.50 : 0.520598
(yolo_env) PS C:\Users\rjrao\Desktop\P_Det>

(yolo_env) PS C:\Users\rjrao\Desktop\P_Det> python code_files/main_execution.py --model models/yolo_onnx_models/F
P16_quantized_models/320/yolov8s_320_fp16.onnx --device cpu --input-size 320 --data-root imagedata
FPS: 21.8552
Model size (on disk): 21.35 MiB (22,382,527 bytes)
mAP@IoU=0.50 : 0.614515
(yolo_env) PS C:\Users\rjrao\Desktop\P_Det>

(yolo_env) PS C:\Users\rjrao\Desktop\P_Det> python code_files/main_execution.py --model models/yolo_onnx_models/F
P16_quantized_models/320/yolov8m_320_fp16.onnx --device cpu --input-size 320 --data-root imagedata
FPS: 11.0203
Model size (on disk): 49.45 MiB (51,854,796 bytes)
mAP@IoU=0.50 : 0.663453
(yolo_env) PS C:\Users\rjrao\Desktop\P_Det>

```

1280 input size

```

(yolo_env) PS C:\Users\rjrao\Desktop\P_Det> python code_files/main_execution.py --model models/yolo_onnx_models/F
P16_quantized_models/1280/yolov8n_1280_fp16.onnx --device cpu --input-size 1280 --data-root imagedata
C:\Users\rjrao\Desktop\P_Det\code_files\postprocessing.py:34: RuntimeWarning: overflow encountered in scalar mult
iply
    inter_area = (x2 - x1).clip(min=0) * (y2 - y1).clip(min=0)
C:\Users\rjrao\Desktop\P_Det\code_files\postprocessing.py:66: RuntimeWarning: invalid value encountered in scalar
divide
    return inter_area / union_area
FPS: 5.1509
Model size (on disk): 6.38 MiB (6,688,163 bytes)
mAP@IoU=0.50 : 0.724195

```

```

(yolo_env) PS C:\Users\rjrao\Desktop\P_Det> python code_files/main_execution.py --model models/yolo_onnx_models/F
P16_quantized_models/1280/yolov8s_1280_fp16.onnx --device cpu --input-size 1280 --data-root imagedata
C:\Users\rjrao\Desktop\P_Det\code_files\postprocessing.py:34: RuntimeWarning: overflow encountered in scalar mult
iply
    inter_area = (x2 - x1).clip(min=0) * (y2 - y1).clip(min=0)
C:\Users\rjrao\Desktop\P_Det\code_files\postprocessing.py:66: RuntimeWarning: invalid value encountered in scalar
divide
    return inter_area / union_area
FPS: 2.2968
Model size (on disk): 21.65 MiB (22,697,545 bytes)
mAP@IoU=0.50 : 0.759581

(yolo_env) PS C:\Users\rjrao\Desktop\P_Det> python code_files/main_execution.py --model models/yolo_onnx_models/F
P16_quantized_models/1280/yolov8m_1280_fp16.onnx --device cpu --input-size 1280 --data-root imagedata
C:\Users\rjrao\Desktop\P_Det\code_files\postprocessing.py:34: RuntimeWarning: overflow encountered in scalar mult
iply
    inter_area = (x2 - x1).clip(min=0) * (y2 - y1).clip(min=0)
C:\Users\rjrao\Desktop\P_Det\code_files\postprocessing.py:66: RuntimeWarning: invalid value encountered in scalar
divide
    return inter_area / union_area
FPS: 0.9149
Model size (on disk): 49.75 MiB (52,169,814 bytes)
mAP@IoU=0.50 : 0.755405

```

INT8 Static Quantized (CPU1)

320 input size

```

(yolo_env) PS C:\Users\rjrao\Desktop\P_Det> python code_files/main_execution.py --model models/yolo_onnx_models/I
NT8_static_quantized_models/320/yolov8n_320_static_quantized.onnx --device cpu --input-size 320 --data-root imag
data
FPS: 16.1451
Model size (on disk): 3.31 MiB (3,474,227 bytes)
mAP@IoU=0.50 : 0.497935

NT8_static_quantized_models/320/yolov8s_320_static_quantized.onnx --device cpu --input-size 320 --data-root image
data
FPS: 8.5828
Model size (on disk): 11.02 MiB (11,554,684 bytes)
mAP@IoU=0.50 : 0.564415

(yolo_env) PS C:\Users\rjrao\Desktop\P_Det> python code_files/main_execution.py --model models/yolo_onnx_models/I
NT8_static_quantized_models/320/yolov8m_320_static_quantized.onnx --device cpu --input-size 320 --data-root image
data
FPS: 8.2609
Model size (on disk): 25.22 MiB (26,448,568 bytes)
mAP@IoU=0.50 : 0.633234
(yolo_env) PS C:\Users\rjrao\Desktop\P_Det>

```

640 input size

```

(yolo_env) PS C:\Users\rjrao\Desktop\P_Det> python code_files/main_execution.py --model models/yolo_onnx_models/I
NT8_static_quantized_models/640/yolov8n_640_static_quantized.onnx --device cpu --input-size 640 --data-root image
data
FPS: 15.5196
Model size (on disk): 3.43 MiB (3,600,269 bytes)
mAP@IoU=0.50 : 0.653009

```

```
(yolo_env) PS C:\Users\rjrao\Desktop\P_Det> python code_files/main_execution.py --model models/yolo_onnx_models/INT8_static_quantized_models/640/yolov8s_640_static_quantized.onnx --device cpu --input-size 640 --data-root image data
FPS: 5.7725
Model size (on disk): 11.14 MiB (11,680,726 bytes)
mAP@IoU=0.50 : 0.691776
```

```
mAP@IoU=0.50 : 0.691776
(yolo_env) PS C:\Users\rjrao\Desktop\P_Det> python code_files/main_execution.py --model models/yolo_onnx_models/INT8_static_quantized_models/640/yolov8m_640_static_quantized.onnx --device cpu --input-size 640 --data-root image data
FPS: 2.6777
Model size (on disk): 25.34 MiB (26,574,624 bytes)
mAP@IoU=0.50 : 0.713427
```

1280 input size

```
(yolo_env) PS C:\Users\rjrao\Desktop\P_Det> python code_files/main_execution.py --model models/yolo_onnx_models/INT8_static_quantized_models/1280/yolov8n_1280_static_quantized.onnx --device cpu --input-size 1280 --data-root imagedata
FPS: 4.0739
Model size (on disk): 3.91 MiB (4,104,411 bytes)
mAP@IoU=0.50 : 0.704670
```

```
(yolo_env) PS C:\Users\rjrao\Desktop\P_Det> python code_files/main_execution.py --model models/yolo_onnx_models/INT8_static_quantized_models/1280/yolov8s_1280_static_quantized.onnx --device cpu --input-size 1280 --data-root imagedata
FPS: 1.5106
Model size (on disk): 11.62 MiB (12,184,868 bytes)
mAP@IoU=0.50 : 0.733785
```

```
(yolo_env) PS C:\Users\rjrao\Desktop\P_Det> python code_files/main_execution.py --model models/yolo_onnx_models/INT8_static_quantized_models/1280/yolov8m_1280_static_quantized.onnx --device cpu --input-size 1280 --data-root imagedata
FPS: 0.7107
Model size (on disk): 25.82 MiB (27,078,806 bytes)
mAP@IoU=0.50 : 0.759759
(yolo_env) PS C:\Users\rjrao\Desktop\P_Det>
```

INT8 Dynamic quantized (CPU1)

320 input size

```
(yolo_env) PS C:\Users\rjrao\Desktop\P_Det> python code_files/main_execution.py --model models\yolo_onnx_models\INT8_dynamic_quantized_models\320\yolov8n_320_dynamic_quantized.onnx --device cpu --input-size 320 --data-root imagedata
FPS: 66.2173
device=cpu
Model size (on disk): 12.14 MiB (12,725,187 bytes)
mAP@IoU=0.50 : 0.521438
```

```
(yolo_env) PS C:\Users\rjrao\Desktop\P_Det> python code_files/main_execution.py --model models\yolo_onnx_models\INT8_dynamic_quantized_models\320\yolov8s_320_dynamic_quantized.onnx --device cpu --input-size 320 --data-root imagedata
FPS: 12.2797
device=cpu
Model size (on disk): 42.67 MiB (44,744,007 bytes)
mAP@IoU=0.50 : 0.614228
```

```
(yolo_env) PS C:\Users\rjrao\Desktop\P_Det> python code_files/main_execution.py --model models\yolo_onnx_models\INT8_dynamic_quantized_models\320\yolov8m_320_dynamic_quantized.onnx --device cpu --input-size 320 --data-root imagedata
FPS: 13.7941
device=cpu
Model size (on disk): 98.88 MiB (103,683,732 bytes)
mAP@IoU=0.50 : 0.660906
```

640 input size

```
(yolo_env) PS C:\Users\rjrao\Desktop\P_Det> python code_files/main_execution.py --model models\yolo_onnx_models\INT8_dynamic_quantized_models\640\yolov8n_640_dynamic_quantized.onnx --device cpu --input-size 640 --data-root imagedata
FPS: 15.0983
device=cpu
Model size (on disk): 12.26 MiB (12,851,269 bytes)
mAP@IoU=0.50 : 0.679877

(yolo_env) PS C:\Users\rjrao\Desktop\P_Det> python code_files/main_execution.py --model models\yolo_onnx_models\INT8_dynamic_quantized_models\640\yolov8s_640_dynamic_quantized.onnx --device cpu --input-size 640 --data-root imagedata
FPS: 9.8318
device=cpu
Model size (on disk): 42.79 MiB (44,870,089 bytes)
mAP@IoU=0.50 : 0.707885

(yolo_env) PS C:\Users\rjrao\Desktop\P_Det> python code_files/main_execution.py --model models\yolo_onnx_models\INT8_dynamic_quantized_models\640\yolov8m_640_dynamic_quantized.onnx --device cpu --input-size 640 --data-root imagedata
FPS: 3.6132
device=cpu
Model size (on disk): 99.00 MiB (103,809,842 bytes)
mAP@IoU=0.50 : 0.723569
```

1280 input size

```
(yolo_env) PS C:\Users\rjrao\Desktop\P_Det> python code_files/main_execution.py --model models\yolo_onnx_models\INT8_dynamic_quantized_models\1280\yolov8n_1280_dynamic_quantized.onnx --device cpu --input-size 1280 --data-root imagedata
FPS: 5.6409
device=cpu
Model size (on disk): 12.74 MiB (13,355,544 bytes)
mAP@IoU=0.50 : 0.725776

(yolo_env) PS C:\Users\rjrao\Desktop\P_Det> python code_files/main_execution.py --model models\yolo_onnx_models\INT8_dynamic_quantized_models\1280\yolov8s_1280_dynamic_quantized.onnx --device cpu --input-size 1280 --data-root imagedata
FPS: 2.4696
device=cpu
Model size (on disk): 43.27 MiB (45,374,364 bytes)
mAP@IoU=0.50 : 0.760904

(yolo_env) PS C:\Users\rjrao\Desktop\P_Det> python code_files/main_execution.py --model models\yolo_onnx_models\INT8_dynamic_quantized_models\1280\yolov8m_1280_dynamic_quantized.onnx --device cpu --input-size 1280 --data-root imagedata
FPS: 0.7726
device=cpu
Model size (on disk): 99.48 MiB (104,314,197 bytes)
mAP@IoU=0.50 : 0.757325
```

FP32 (GPU1)

320

```
(yolo_env) PS C:\Users\rjrao\Desktop\P_Det> python code_files/main_execution.py --model models\yolo_onnx_models\FP32_onnx_models\320\yolov8n_320.onnx --device gpu --input-size 320 --data-root imagedata
FPS: 68.4923
device=gpu
Model size (on disk): 12.14 MiB (12,725,005 bytes)
mAP@IoU=0.50 : 0.521454
(yolo_env) PS C:\Users\rjrao\Desktop\P_Det>
```

```
(yolo_env) PS C:\Users\rjrao\Desktop\P_Det> python code_files/main_execution.py --model models\yolo_onnx_models\FP32_onnx_m
odels\320\yolov8s_320.onnx --device gpu --input-size 320 --data-root imagedata
FPS: 57.3142
device=gpu
Model size (on disk): 42.67 MiB (44,743,799 bytes)
mAP@IoU=0.50 : 0.614213

(yolo_env) PS C:\Users\rjrao\Desktop\P_Det> python code_files/main_execution.py --model models\yolo_onnx_models\FP32_onnx_m
odels\320\yolov8m_320.onnx --device gpu --input-size 320 --data-root imagedata
FPS: 28.9642
device=gpu
Model size (on disk): 98.88 MiB (103,683,476 bytes)
mAP@IoU=0.50 : 0.660919
(yolo_env) PS C:\Users\rjrao\Desktop\P_Det> █
```

640

```
(yolo_env) PS C:\Users\rjrao\Desktop\P_Det> python code_files/main_execution.py --model models\yolo_onnx_models\FP32_onnx_m
odels\640\yolov8n_640.onnx --device gpu --input-size 640 --data-root imagedata
FPS: 39.4660
device=gpu
Model size (on disk): 12.26 MiB (12,851,087 bytes)
mAP@IoU=0.50 : 0.679865

(yolo_env) PS C:\Users\rjrao\Desktop\P_Det> python code_files/main_execution.py --model models\yolo_onnx_models\FP32_onnx_m
odels\640\yolov8s_640.onnx --device gpu --input-size 640 --data-root imagedata
FPS: 25.7842
device=gpu
Model size (on disk): 42.79 MiB (44,869,881 bytes)
mAP@IoU=0.50 : 0.707891

(yolo_env) PS C:\Users\rjrao\Desktop\P_Det> python code_files/main_execution.py --model models\yolo_onnx_models\FP32_onnx_m
odels\640\yolov8m_640.onnx --device gpu --input-size 640 --data-root imagedata
FPS: 13.9254
device=gpu
Model size (on disk): 99.00 MiB (103,809,586 bytes)
mAP@IoU=0.50 : 0.723554
(yolo_env) PS C:\Users\rjrao\Desktop\P_Det> █
```

1280

```
mAP@IoU=0.50 : 0.723554
(yolo_env) PS C:\Users\rjrao\Desktop\P_Det> python code_files/main_execution.py --model models\yolo_onnx_models\FP32_onnx_m
odels\1280\yolov8n_1280.onnx --device gpu --input-size 1280 --data-root imagedata
FPS: 14.8924
device=gpu
Model size (on disk): 12.74 MiB (13,355,362 bytes)
mAP@IoU=0.50 : 0.725816

(yolo_env) PS C:\Users\rjrao\Desktop\P_Det> python code_files/main_execution.py --model models\yolo_onnx_models\FP32_onnx_m
odels\1280\yolov8s_1280.onnx --device gpu --input-size 1280 --data-root imagedata
FPS: 9.2814
device=gpu
Model size (on disk): 43.27 MiB (45,374,156 bytes)
mAP@IoU=0.50 : 0.760830
python code_files/main_execution.py --model models\yolo_onnx_models\FP32_onnx_m

(yolo_env) PS C:\Users\rjrao\Desktop\P_Det> python code_files/main_execution.py --model models\yolo_onnx_models\FP32_onnx_m
odels\1280\yolov8m_1280.onnx --device gpu --input-size 1280 --data-root imagedata
FPS: 5.1672
device=gpu
Model size (on disk): 99.48 MiB (104,313,941 bytes)
mAP@IoU=0.50 : 0.757285
(yolo_env) PS C:\Users\rjrao\Desktop\P_Det> █
```

INT8 Static quantization (GPU1)

320

```
(yolo_env) PS C:\Users\rjrao\Desktop\P_Det> python code_files/main_execution.py --model models\yolo_onnx_models\INT8_static_quantized_models\320\yolov8n_320_static_quantized.onnx --device gpu --input-size 320 --data-root imagedata
2026-05-03 22:16:42.9800676 [W:onnxruntime:, transformer_memcpy.cc:111 onnxruntime::MemcpyTransformer::ApplyImpl] 63 Memcpy nodes are added to the graph main_graph for CUDAEExecutionProvider. It might have negative impact on performance (including unable to run CUDA graph). Set session_options.log_severity_level=1 to see the detail logs before this message.
FPS: 52.0796
device=gpu
Model size (on disk): 3.31 MiB (3,474,227 bytes)
mAP@IoU=0.50 : 0.491260
(yolo_env) PS C:\Users\rjrao\Desktop\P_Det>
```

```
(yolo_env) PS C:\Users\rjrao\Desktop\P_Det> python code_files/main_execution.py --model models\yolo_onnx_models\INT8_static_quantized_models\320\yolov8s_320_static_quantized.onnx --device gpu --input-size 320 --data-root imagedata
2026-05-03 22:17:08.8398341 [W:onnxruntime:, transformer_memcpy.cc:111 onnxruntime::MemcpyTransformer::ApplyImpl] 63 Memcpy nodes are added to the graph main_graph for CUDAEExecutionProvider. It might have negative impact on performance (including unable to run CUDA graph). Set session_options.log_severity_level=1 to see the detail logs before this message.
FPS: 42.9500
device=gpu
Model size (on disk): 11.02 MiB (11,554,684 bytes)
mAP@IoU=0.50 : 0.564701
(yolo_env) PS C:\Users\rjrao\Desktop\P_Det>
```

```
(yolo_env) PS C:\Users\rjrao\Desktop\P_Det> python code_files/main_execution.py --model models\yolo_onnx_models\INT8_static_quantized_models\320\yolov8m_320_static_quantized.onnx --device gpu --input-size 320 --data-root imagedata
2026-05-03 22:17:23.7182927 [W:onnxruntime:, transformer_memcpy.cc:111 onnxruntime::MemcpyTransformer::ApplyImpl] 83 Memcpy nodes are added to the graph main_graph for CUDAEExecutionProvider. It might have negative impact on performance (including unable to run CUDA graph). Set session_options.log_severity_level=1 to see the detail logs before this message.
FPS: 24.4121
device=gpu
Model size (on disk): 25.22 MiB (26,448,568 bytes)
mAP@IoU=0.50 : 0.626899
(yolo_env) PS C:\Users\rjrao\Desktop\P_Det>
```

640

```
mAP@IoU=0.50 : 0.626899
(yolo_env) PS C:\Users\rjrao\Desktop\P_Det> python code_files/main_execution.py --model models\yolo_onnx_models\INT8_static_quantized_models\640\yolov8n_640_static_quantized.onnx --device gpu --input-size 640 --data-root imagedata
2026-05-03 22:19:16.5935512 [W:onnxruntime:, transformer_memcpy.cc:111 onnxruntime::MemcpyTransformer::ApplyImpl] 63 Memcpy nodes are added to the graph main_graph for CUDAEExecutionProvider. It might have negative impact on performance (including unable to run CUDA graph). Set session_options.log_severity_level=1 to see the detail logs before this message.
FPS: 30.5967
device=gpu
Model size (on disk): 3.43 MiB (3,600,269 bytes)
mAP@IoU=0.50 : 0.658708
(yolo_env) PS C:\Users\rjrao\Desktop\P_Det>
```

```
(yolo_env) PS C:\Users\rjrao\Desktop\P_Det> python code_files/main_execution.py --model models\yolo_onnx_models\INT8_static_quantized_models\640\yolov8s_640_static_quantized.onnx --device gpu --input-size 640 --data-root imagedata
2026-05-03 22:19:29.6263499 [W:onnxruntime:, transformer_memcpy.cc:111 onnxruntime::MemcpyTransformer::ApplyImpl] 63 Memcpy nodes are added to the graph main_graph for CUDAEExecutionProvider. It might have negative impact on performance (including unable to run CUDA graph). Set session_options.log_severity_level=1 to see the detail logs before this message.
FPS: 22.3620
device=gpu
Model size (on disk): 11.14 MiB (11,680,726 bytes)
mAP@IoU=0.50 : 0.684866
(yolo_env) PS C:\Users\rjrao\Desktop\P_Det>
```

```
mAP@IoU=0.50 : 0.684866
(yolo_env) PS C:\Users\rjrao\Desktop\P_Det> python code_files/main_execution.py --model models\yolo_onnx_models\INT8_static_quantized_models\640\yolov8m_640_static_quantized.onnx --device gpu --input-size 640 --data-root imagedata
2026-05-03 22:19:40.5292893 [W:onnxruntime:, transformer_memcpy.cc:111 onnxruntime::MemcpyTransformer::ApplyImpl] 83 Memcpy nodes are added to the graph main_graph for CUDAEExecutionProvider. It might have negative impact on performance (including unable to run CUDA graph). Set session_options.log_severity_level=1 to see the detail logs before this message.
FPS: 11.8089
device=gpu
Model size (on disk): 25.34 MiB (26,574,624 bytes)
mAP@IoU=0.50 : 0.708243
(yolo_env) PS C:\Users\rjrao\Desktop\P_Det>
```

1280

```
• _quantized_models\1280\yolov8n_1280_static_quantized.onnx --device gpu --input-size 1280 --data-root imagedata
2026-05-03 22:20:23.6149767 [W:onnxruntime:, transformer_memcpy.cc:111 onnxruntime::MemcpyTransformer::ApplyImpl] 63 Memcpy
nodes are added to the graph main_graph for CUDAEExecutionProvider. It might have negative impact on performance (including
unable to run CUDA graph). Set session_options.log_severity_level=1 to see the detail logs before this message.
FPS: 11.5545
device=gpu
Model size (on disk): 3.91 MiB (4,104,411 bytes)
mAP@IoU=0.50 : 0.710115
(yolo_env) PS C:\Users\rjrao\Desktop\P_Det>

• (yolo_env) PS C:\Users\rjrao\Desktop\P_Det> python code_files/main_execution.py --model models\yolo_onnx_models\INT8_static
_quantized_models\1280\yolov8s_1280_static_quantized.onnx --device gpu --input-size 1280 --data-root imagedata
2026-05-03 22:20:49.3277053 [W:onnxruntime:, transformer_memcpy.cc:111 onnxruntime::MemcpyTransformer::ApplyImpl] 63 Memcpy
nodes are added to the graph main_graph for CUDAEExecutionProvider. It might have negative impact on performance (including
unable to run CUDA graph). Set session_options.log_severity_level=1 to see the detail logs before this message.
FPS: 8.1129
device=gpu
Model size (on disk): 11.62 MiB (12,184,868 bytes)
mAP@IoU=0.50 : 0.731770

• (yolo_env) PS C:\Users\rjrao\Desktop\P_Det> python code_files/main_execution.py --model models\yolo_onnx_models\INT8_static
_quantized_models\1280\yolov8m_1280_static_quantized.onnx --device gpu --input-size 1280 --data-root imagedata
2026-05-03 22:21:09.9473603 [W:onnxruntime:, transformer_memcpy.cc:111 onnxruntime::MemcpyTransformer::ApplyImpl] 83 Memcpy
nodes are added to the graph main_graph for CUDAEExecutionProvider. It might have negative impact on performance (including
unable to run CUDA graph). Set session_options.log_severity_level=1 to see the detail logs before this message.
FPS: 4.0203
device=gpu
Model size (on disk): 25.82 MiB (27,078,806 bytes)
mAP@IoU=0.50 : 0.747494
```

INT8 dynamic quantization (GPU1)

320 input size

```
(yolo_env) PS C:\Users\rjrao\Desktop\P_Det> python code_files/main_execution.py --model models\yolo_onnx_models\INT8_dynamic_quantized
_models\320\yolov8n_320_dynamic_quantized.onnx --device gpu --input-size 320 --data-root imagedata
FPS: 68.7856
device=gpu
Model size (on disk): 12.14 MiB (12,725,187 bytes)
mAP@IoU=0.50 : 0.521454

(yolo_env) PS C:\Users\rjrao\Desktop\P_Det> python code_files/main_execution.py --model models\yolo_onnx_models\INT8_dynamic_quantized
_models\320\yolov8s_320_dynamic_quantized.onnx --device gpu --input-size 320 --data-root imagedata
FPS: 48.6796
device=gpu
Model size (on disk): 42.67 MiB (44,744,007 bytes)
mAP@IoU=0.50 : 0.614213

(yolo_env) PS C:\Users\rjrao\Desktop\P_Det> python code_files/main_execution.py --model models\yolo_onnx_models\INT8_dynamic_q
_models\320\yolov8m_320_dynamic_quantized.onnx --device gpu --input-size 320 --data-root imagedata
FPS: 29.7171
device=gpu
Model size (on disk): 98.88 MiB (103,683,732 bytes)
mAP@IoU=0.50 : 0.660919
```

640 input size

```
(yolo_env) PS C:\Users\rjrao\Desktop\P_Det> python code_files/main_execution.py --model models\yolo_onnx_models\INT8_dynamic_quantized
_models\640\yolov8n_640_dynamic_quantized.onnx --device gpu --input-size 640 --data-root imagedata
FPS: 38.7926
device=gpu
Model size (on disk): 12.26 MiB (12,851,269 bytes)
mAP@IoU=0.50 : 0.679865
```

```
(yolo_env) PS C:\Users\rjrao\Desktop\P_Det> python code_files/main_execution.py --model models\yolo_onnx_models\INT8_dynamic_quantized_models\640\yolov8s_640_dynamic_quantized.onnx --device gpu --input-size 640 --data-root imagedata
FPS: 25.9110
device=gpu
Model size (on disk): 42.79 MiB (44,870,089 bytes)
mAP@IoU=0.50 : 0.707891
```

```
(yolo_env) PS C:\Users\rjrao\Desktop\P_Det> python code_files/main_execution.py --model models\yolo_onnx_models\INT8_dynamic_quantized_models\640\yolov8m_640_dynamic_quantized.onnx --device gpu --input-size 640 --data-root imagedata
FPS: 16.7324
device=gpu
Model size (on disk): 99.00 MiB (103,809,842 bytes)
mAP@IoU=0.50 : 0.723554
```

1280 input size

```
(yolo_env) PS C:\Users\rjrao\Desktop\P_Det> python code_files/main_execution.py --model models\yolo_onnx_models\INT8_dynamic_quantized_models\1280\yolov8n_1280_dynamic_quantized.onnx --device gpu --input-size 1280 --data-root imagedata
FPS: 11.3084
device=gpu
Model size (on disk): 12.74 MiB (13,355,544 bytes)
mAP@IoU=0.50 : 0.725816
```

```
(yolo_env) PS C:\Users\rjrao\Desktop\P_Det> python code_files/main_execution.py --model models\yolo_onnx_models\INT8_dynamic_quantized_models\1280\yolov8s_1280_dynamic_quantized.onnx --device gpu --input-size 1280 --data-root imagedata
FPS: 9.8726
device=gpu
Model size (on disk): 43.27 MiB (45,374,364 bytes)
mAP@IoU=0.50 : 0.760830
```

```
(yolo_env) PS C:\Users\rjrao\Desktop\P_Det> python code_files/main_execution.py --model models\yolo_onnx_models\INT8_dynamic_quantized_models\1280\yolov8m_1280_dynamic_quantized.onnx --device gpu --input-size 1280 --data-root imagedata
FPS: 5.1728
device=gpu
Model size (on disk): 99.48 MiB (104,314,197 bytes)
mAP@IoU=0.50 : 0.757285
```

FP16 quantization (GPU1)

320 input size

```
(yolo_env) PS C:\Users\rjrao\Desktop\P_Det> python code_files/main_execution.py --model models\yolo_onnx_models\FP16_quantized_models\320\yolov8n_320_fp16.onnx --device gpu --input-size 320 --data-root imagedata
FPS: 51.4656
device=gpu
Model size (on disk): 6.08 MiB (6,373,145 bytes)
mAP@IoU=0.50 : 0.520139
```

```
(yolo_env) PS C:\Users\rjrao\Desktop\P_Det> python code_files/main_execution.py --model models\yolo_onnx_models\FP16_quantized_models\320\yolov8s_320_fp16.onnx --device gpu --input-size 320 --data-root imagedata
FPS: 50.7065
device=gpu
Model size (on disk): 21.35 MiB (22,382,527 bytes)
mAP@IoU=0.50 : 0.614227
```

```
(yolo_env) PS C:\Users\rjrao\Desktop\P_Det> python code_files/main_execution.py --model models\yolo_onnx_models\FP16_quantized_models\320\yolov8m_320_fp16.onnx --device gpu --input-size 320 --data-root imagedata
FPS: 28.2521
device=gpu
Model size (on disk): 49.45 MiB (51,854,796 bytes)
mAP@IoU=0.50 : 0.659276
```

640 input size

```
(yolo_env) PS C:\Users\rjrao\Desktop\P_Det> python code_files/main_execution.py --model models\yolo_onnx_models\FP16_quantized_models\640\yolov8n_640_fp16.onnx --device gpu --input-size 640 --data-root imagedata
FPS: 19.7382
device=gpu
Model size (on disk): 6.14 MiB (6,436,157 bytes)
mAP@IoU=0.50 : 0.679815

(yolo_env) PS C:\Users\rjrao\Desktop\P_Det> python code_files/main_execution.py --model models\yolo_onnx_models\FP16_quantized_models\640\yolov8s_640_fp16.onnx --device gpu --input-size 640 --data-root imagedata
FPS: 28.1420
device=gpu
Model size (on disk): 21.41 MiB (22,445,539 bytes)
mAP@IoU=0.50 : 0.709141

(yolo_env) PS C:\Users\rjrao\Desktop\P_Det> python code_files/main_execution.py --model models\yolo_onnx_models\FP16_quantized_models\640\yolov8m_640_fp16.onnx --device gpu --input-size 640 --data-root imagedata
FPS: 19.2476
device=gpu
Model size (on disk): 49.51 MiB (51,917,808 bytes)
mAP@IoU=0.50 : 0.723527
```

1280 input size

```
(yolo_env) PS C:\Users\rjrao\Desktop\P_Det> python code_files/main_execution.py --model models\yolo_onnx_models\FP16_quantized_models\1280\yolov8n_1280_fp16.onnx --device gpu --input-size 1280 --data-root imagedata
FPS: 10.5133
device=gpu
Model size (on disk): 6.38 MiB (6,688,163 bytes)
mAP@IoU=0.50 : 0.725673

(yolo_env) PS C:\Users\rjrao\Desktop\P_Det> python code_files/main_execution.py --model models\yolo_onnx_models\FP16_quantized_models\1280\yolov8s_1280_fp16.onnx --device gpu --input-size 1280 --data-root imagedata
FPS: 7.8868
device=gpu
Model size (on disk): 21.65 MiB (22,697,545 bytes)
mAP@IoU=0.50 : 0.761154

(yolo_env) PS C:\Users\rjrao\Desktop\P_Det> python code_files/main_execution.py --model models\yolo_onnx_models\FP16_quantized_models\1280\yolov8m_1280_fp16.onnx --device gpu --input-size 1280 --data-root imagedata
FPS: 5.1841
device=gpu
Model size (on disk): 49.75 MiB (52,169,814 bytes)
mAP@IoU=0.50 : 0.757309
(yolo_env) PS C:\Users\rjrao\Desktop\P_Det>
```

FP32 (GPU2)

320 input size

```
-cudf-cu12-11.4.1.4 nvdata-cuda-cu12-10.3.10.19 nvtools-nvjitlink-cu12-12.9.86
(yolo_quant) PS F:\Final_Lattice_Data_Structure\ActiveMixedMaterialDataset\Suraj_Code_Files\P_Det> python .\code_files\main_execution.py --model "models\yolo_onnx_models\FP32_onnx_models\320\yolov8n_320.onnx" --device gpu --input-size 320 --data-root ".\imagedata"
FPS: 69.8035
device=gpu
Model size (on disk): 12.14 MiB (12,725,005 bytes)
mAP@IoU=0.50 : 0.518815
(yolo_quant) PS F:\Final_Lattice_Data_Structure\ActiveMixedMaterialDataset\Suraj_Code_Files\P_Det>

(yolo_quant) PS F:\Final_Lattice_Data_Structure\ActiveMixedMaterialDataset\Suraj_Code_Files\P_Det> python .\code_files\main_execution.py --model "models\yolo_onnx_models\FP32_onnx_models\320\yolov8s_320.onnx" --device gpu --input-size 320 --data-root ".\imagedata"
FPS: 67.1662
device=gpu
Model size (on disk): 42.67 MiB (44,743,799 bytes)
mAP@IoU=0.50 : 0.614202

(yolo_quant) PS F:\Final_Lattice_Data_Structure\ActiveMixedMaterialDataset\Suraj_Code_Files\P_Det> python .\code_files\main_execution.py --model "models\yolo_onnx_models\FP32_onnx_models\320\yolov8m_320.onnx" --device gpu --input-size 320 --data-root ".\imagedata"
FPS: 56.2887
device=gpu
Model size (on disk): 98.88 MiB (103,683,476 bytes)
mAP@IoU=0.50 : 0.660910
(yolo_quant) PS F:\Final_Lattice_Data_Structure\ActiveMixedMaterialDataset\Suraj_Code_Files\P_Det>
```

640 input size

```
(yolo_quant) PS F:\Final_Lattice_Data_Structure\ActiveMixedMaterialDataset\Suraj_Code_Files\P_Det> python .\code_files\main_execution.py --model "models\yolo_onnx_models\FP32_onnx_models\640\yolov8n_640.onnx" --device gpu --input-size 640 --data-root ".\imagedata"
FPS: 43.7517
device=gpu
Model size (on disk): 12.26 MiB (12,851,087 bytes)
mAP@IoU=0.50 : 0.679870
(yolo_quant) PS F:\Final_Lattice_Data_Structure\ActiveMixedMaterialDataset\Suraj_Code_Files\P_Det>

(yolo_quant) PS F:\Final_Lattice_Data_Structure\ActiveMixedMaterialDataset\Suraj_Code_Files\P_Det> python .\code_files\main_execution.py --model "models\yolo_onnx_models\FP32_onnx_models\640\yolov8s_640.onnx" --device gpu --input-size 640 --data-root ".\imagedata"
FPS: 40.6903
device=gpu
Model size (on disk): 42.79 MiB (44,869,881 bytes)
mAP@IoU=0.50 : 0.707891
(yolo_quant) PS F:\Final_Lattice_Data_Structure\ActiveMixedMaterialDataset\Suraj_Code_Files\P_Det>

(yolo_quant) PS F:\Final_Lattice_Data_Structure\ActiveMixedMaterialDataset\Suraj_Code_Files\P_Det> python .\code_files\main_execution.py --model "models\yolo_onnx_models\FP32_onnx_models\640\yolov8m_640.onnx" --device gpu --input-size 640 --data-root ".\imagedata"
FPS: 35.0169
device=gpu
Model size (on disk): 99.00 MiB (103,809,586 bytes)
mAP@IoU=0.50 : 0.723554
(yolo_quant) PS F:\Final_Lattice_Data_Structure\ActiveMixedMaterialDataset\Suraj_Code_Files\P_Det>
```

1280 input size

```
(yolo_quant) PS F:\Final_Lattice_Data_Structure\ActiveMixedMaterialDataset\Suraj_Code_Files\P_Det> python .\code_files\main_execution.py --model "models\yolo_onnx_models\FP32_onnx_models\1280\yolov8n_1280.onnx" --device gpu --input-size 1280 --data-root ".\imagedata"
FPS: 17.9713
device=gpu
Model size (on disk): 12.74 MiB (13,355,362 bytes)
mAP@IoU=0.50 : 0.725816
(yolo_quant) PS F:\Final_Lattice_Data_Structure\ActiveMixedMaterialDataset\Suraj_Code_Files\P_Det>

(yolo_quant) PS F:\Final_Lattice_Data_Structure\ActiveMixedMaterialDataset\Suraj_Code_Files\P_Det> python .\code_files\main_execution.py --model "models\yolo_onnx_models\FP32_onnx_models\1280\yolov8s_1280.onnx" --device gpu --input-size 1280 --data-root ".\imagedata"
FPS: 15.0469
device=gpu
Model size (on disk): 43.27 MiB (45,374,156 bytes)
mAP@IoU=0.50 : 0.760824
(yolo_quant) PS F:\Final_Lattice_Data_Structure\ActiveMixedMaterialDataset\Suraj_Code_Files\P_Det>

(yolo_quant) PS F:\Final_Lattice_Data_Structure\ActiveMixedMaterialDataset\Suraj_Code_Files\P_Det> python .\code_files\main_execution.py --model "models\yolo_onnx_models\FP32_onnx_models\1280\yolov8m_1280.onnx" --device gpu --input-size 1280 --data-root ".\imagedata"
FPS: 13.1702
device=gpu
Model size (on disk): 99.48 MiB (104,313,941 bytes)
mAP@IoU=0.50 : 0.757285
(yolo_quant) PS F:\Final_Lattice_Data_Structure\ActiveMixedMaterialDataset\Suraj_Code_Files\P_Det>
```

FP16 quantization (GPU2)

320 input size

```
(yolo_quant) PS F:\Final_Lattice_Data_Structure\ActiveMixedMaterialDataset\Suraj_Code_Files\P_Det> python .\code_files\main_execution.py --model "models\yolo_onnx_models\FP16_quantized_models\320\yolov8n_320_fp16.onnx" --device gpu --input-size 320 --data-root ".\imagedata"
FPS: 63.8759
device=gpu
Model size (on disk): 6.08 MiB (6,373,145 bytes)
mAP@IoU=0.50 : 0.520139
(yolo_quant) PS F:\Final_Lattice_Data_Structure\ActiveMixedMaterialDataset\Suraj_Code_Files\P_Det>

(yolo_quant) PS F:\Final_Lattice_Data_Structure\ActiveMixedMaterialDataset\Suraj_Code_Files\P_Det> python .\code_files\main_execution.py --model "models\yolo_onnx_models\FP16_quantized_models\320\yolov8s_320_fp16.onnx" --device gpu --input-size 320 --data-root ".\imagedata"
FPS: 57.6318
device=gpu
Model size (on disk): 21.35 MiB (22,382,527 bytes)
mAP@IoU=0.50 : 0.614337
(yolo_quant) PS F:\Final_Lattice_Data_Structure\ActiveMixedMaterialDataset\Suraj_Code_Files\P_Det>

(yolo_quant) PS F:\Final_Lattice_Data_Structure\ActiveMixedMaterialDataset\Suraj_Code_Files\P_Det> python .\code_files\main_execution.py --model "models\yolo_onnx_models\FP16_quantized_models\320\yolov8m_320_fp16.onnx" --device gpu --input-size 320 --data-root ".\imagedata"
FPS: 50.3925
device=gpu
Model size (on disk): 49.45 MiB (51,854,796 bytes)
mAP@IoU=0.50 : 0.660755
(yolo_quant) PS F:\Final_Lattice_Data_Structure\ActiveMixedMaterialDataset\Suraj_Code_Files\P_Det>
```

640 input size

```
(yolo_quant) PS F:\Final_Lattice_Data_Structure\ActiveMixedMaterialDataset\Suraj_Code_Files\P_Det> python .\code_files\main_execution.py --model "models\yolo_onnx_models\FP16_quantized_models\640\yolov8n_640_fp16.onnx" --device gpu --input-size 640 --data-root ".\imagedata"
FPS: 34.2168
device=gpu
Model size (on disk): 6.14 MiB (6,436,157 bytes)
mAP@IoU=0.50 : 0.679815

(yolo_quant) PS F:\Final_Lattice_Data_Structure\ActiveMixedMaterialDataset\Suraj_Code_Files\P_Det> python .\code_files\main_execution.py --model "models\yolo_onnx_models\FP16_quantized_models\640\yolov8s_640_fp16.onnx" --device gpu --input-size 640 --data-root ".\imagedata"
FPS: 30.1633
device=gpu
Model size (on disk): 21.41 MiB (22,445,539 bytes)
mAP@IoU=0.50 : 0.709162

(yolo_quant) PS F:\Final_Lattice_Data_Structure\ActiveMixedMaterialDataset\Suraj_Code_Files\P_Det> python .\code_files\main_execution.py --model "models\yolo_onnx_models\FP16_quantized_models\640\yolov8m_640_fp16.onnx" --device gpu --input-size 640 --data-root ".\imagedata"
FPS: 30.4514
device=gpu
Model size (on disk): 49.51 MiB (51,917,808 bytes)
mAP@IoU=0.50 : 0.723568
```

1280 input size

```
(yolo_quant) PS F:\Final_Lattice_Data_Structure\ActiveMixedMaterialDataset\Suraj_Code_Files\P_Det> python .\code_files\main_execution.py --model "models\yolo_onnx_models\FP16_quantized_models\1280\yolov8n_1280_fp16.onnx" --device gpu --input-size 1280 --data-root ".\imagedata"
FPS: 13.2606
device=gpu
Model size (on disk): 6.38 MiB (6,688,163 bytes)
mAP@IoU=0.50 : 0.725643

(yolo_quant) PS F:\Final_Lattice_Data_Structure\ActiveMixedMaterialDataset\Suraj_Code_Files\P_Det> python .\code_files\main_execution.py --model "models\yolo_onnx_models\FP16_quantized_models\1280\yolov8s_1280_fp16.onnx" --device gpu --input-size 1280 --data-root ".\imagedata"
FPS: 12.4568
device=gpu
Model size (on disk): 21.65 MiB (22,697,545 bytes)
mAP@IoU=0.50 : 0.761018

(yolo_quant) PS F:\Final_Lattice_Data_Structure\ActiveMixedMaterialDataset\Suraj_Code_Files\P_Det> python .\code_files\main_execution.py --model "models\yolo_onnx_models\FP16_quantized_models\1280\yolov8m_1280_fp16.onnx" --device gpu --input-size 1280 --data-root ".\imagedata"
FPS: 10.7555
device=gpu
Model size (on disk): 49.75 MiB (52,169,814 bytes)
mAP@IoU=0.50 : 0.757326
```

INT8 Dynamic Quantization (GPU2)

320 input size

```
(yolo_quant) PS F:\Final_Lattice_Data_Structure\ActiveMixedMaterialDataset\Suraj_Code_Files\P_Det> python .\code_files\main_execution.py --model "models\yolo_onnx_models\INT8_dynamic_quantized_models\320\yolov8n_320_dynamic_quantized.onnx" --device gpu --input-size 320 --data-root ".\imagedata"
FPS: 74.1708
device=gpu
Model size (on disk): 12.14 MiB (12,725,187 bytes)
mAP@IoU=0.50 : 0.518815

(yolo_quant) PS F:\Final_Lattice_Data_Structure\ActiveMixedMaterialDataset\Suraj_Code_Files\P_Det> python .\code_files\main_execution.py --model "models\yolo_onnx_models\INT8_dynamic_quantized_models\320\yolov8s_320_dynamic_quantized.onnx" --device gpu --input-size 320 --data-root ".\imagedata"
FPS: 67.5562
device=gpu
Model size (on disk): 42.67 MiB (44,744,007 bytes)
mAP@IoU=0.50 : 0.614202

(yolo_quant) PS F:\Final_Lattice_Data_Structure\ActiveMixedMaterialDataset\Suraj_Code_Files\P_Det> python .\code_files\main_execution.py --model "models\yolo_onnx_models\INT8_dynamic_quantized_models\320\yolov8m_320_dynamic_quantized.onnx" --device gpu --input-size 320 --data-root ".\imagedata"
FPS: 60.9990
device=gpu
Model size (on disk): 98.88 MiB (103,683,732 bytes)
mAP@IoU=0.50 : 0.660910
```

640 input size

```
(yolo_quant) PS F:\Final_Lattice_Data_Structure\ActiveMixedMaterialDataset\Suraj_Code_Files\P_Det> python .\code_files\main_execution.py --model "models\yolo_onnx_models\INT8_dynamic_quantized_models\640\yolov8n_640_dynamic_quantized.onnx" --device gpu --input-size 640 --data-root ".\imagedata"
FPS: 45.9126
device=gpu
Model size (on disk): 12.26 MiB (12,851,269 bytes)
mAP@IoU=0.50 : 0.679870

(yolo_quant) PS F:\Final_Lattice_Data_Structure\ActiveMixedMaterialDataset\Suraj_Code_Files\P_Det> python .\code_files\main_execution.py --model "models\yolo_onnx_models\INT8_dynamic_quantized_models\640\yolov8s_640_dynamic_quantized.onnx" --device gpu --input-size 640 --data-root ".\imagedata"
FPS: 41.3695
device=gpu
Model size (on disk): 42.79 MiB (44,870,889 bytes)
mAP@IoU=0.50 : 0.707891

(yolo_quant) PS F:\Final_Lattice_Data_Structure\ActiveMixedMaterialDataset\Suraj_Code_Files\P_Det> python .\code_files\main_execution.py --model "models\yolo_onnx_models\INT8_dynamic_quantized_models\640\yolov8m_640_dynamic_quantized.onnx" --device gpu --input-size 640 --data-root ".\imagedata"
FPS: 36.8402
device=gpu
Model size (on disk): 99.00 MiB (103,809,842 bytes)
mAP@IoU=0.50 : 0.723554
```

1280 input size

```
(yolo_quant) PS F:\Final_Lattice_Data_Structure\ActiveMixedMaterialDataset\Suraj_Code_Files\P_Det> python .\code_files\main_execution.py --model "models\yolo_onnx_models\INT8_dynamic_quantized_models\1280\yolov8n_1280_dynamic_quantized.onnx" --device gpu --input-size 1280 --data-root ".\imagedata"
FPS: 18.1003
device=gpu
Model size (on disk): 12.74 MiB (13,355,544 bytes)
mAP@IoU=0.50 : 0.725816
(yolo_quant) PS F:\Final_Lattice_Data_Structure\ActiveMixedMaterialDataset\Suraj_Code_Files\P_Det>

(yolo_quant) PS F:\Final_Lattice_Data_Structure\ActiveMixedMaterialDataset\Suraj_Code_Files\P_Det> python .\code_files\main_execution.py --model "models\yolo_onnx_models\INT8_dynamic_quantized_models\1280\yolov8s_1280_dynamic_quantized.onnx" --device gpu --input-size 1280 --data-root ".\imagedata"
FPS: 16.5224
device=gpu
Model size (on disk): 43.27 MiB (45,374,364 bytes)
mAP@IoU=0.50 : 0.760824

(yolo_quant) PS F:\Final_Lattice_Data_Structure\ActiveMixedMaterialDataset\Suraj_Code_Files\P_Det> python .\code_files\main_execution.py --model "models\yolo_onnx_models\INT8_dynamic_quantized_models\1280\yolov8m_1280_dynamic_quantized.onnx" --device gpu --input-size 1280 --data-root ".\imagedata"
FPS: 13.1065
device=gpu
Model size (on disk): 99.48 MiB (104,314,197 bytes)
mAP@IoU=0.50 : 0.757285
(yolo_quant) PS F:\Final_Lattice_Data_Structure\ActiveMixedMaterialDataset\Suraj_Code_Files\P_Det>
```

INT8 static quantization (GPU2)

320 input size

```
(yolo_quant) PS F:\Final_Lattice_Data_Structure\ActiveMixedMaterialDataset\Suraj_Code_Files\P_Det> python .\code_files\main_execution.py --model "models\yolo_onnx_models\INT8_static_quantized_models\320\yolov8s_320_static_quantized.onnx" --device gpu --input-size 320 --data-root ".\imagedata"
2026-05-04 10:53:27.7155876 [W:onnxruntime:, transformer_memcpy.cc:111 onnxruntime:MemcpyTransformer::ApplyImpl] 63 Mempcy nodes are added to the graph main_graph for CUDAEExecutionProvider. It might have negative impact on performance (including unable to run CUDA graph). Set session_options.log_severity_level=1 to see the detail logs before this message.
FPS: 51.9161
device=gpu
Model size (on disk): 11.02 MiB (11,554,684 bytes)
mAP@IoU=0.50 : 0.564533
```

```

(yolo_quant) PS F:\Final_Lattice_Data_Structure\ActiveMixedMaterialDataset\Suraj_Code_Files\P_Det> python .\code_files\main_execution.py --model "models\yolo_onnx_models\INT8_static_quantized_models\320\yolov8n_320_static_quantized.onnx" --device gpu --input-size 320 --data-root ".\imagedata"
2026-05-04 10:53:10.8415642 [W:onnxruntime:, transformer_memcpy.cc:111 onnxruntime::MemcpyTransformer::ApplyImpl] 63 Mempcy nodes are added to the graph main_graph for CUDAEExecutionProvider. It might have negative impact on performance (including unable to run CUDA graph). Set session_options.log_severity_level=1 to see the detail logs before this message.
FPS: 57.1072
device=gpu
Model size (on disk): 3.31 MiB (3,474,227 bytes)
mAP@IoU=0.50 : 0.505161

(yolo_quant) PS F:\Final_Lattice_Data_Structure\ActiveMixedMaterialDataset\Suraj_Code_Files\P_Det> python .\code_files\main_execution.py --model "models\yolo_onnx_models\INT8_static_quantized_models\320\yolov8m_320_static_quantized.onnx" --device gpu --input-size 320 --data-root ".\imagedata"
2026-05-04 10:53:45.9383275 [W:onnxruntime:, transformer_memcpy.cc:111 onnxruntime::MemcpyTransformer::ApplyImpl] 63 Mempcy nodes are added to the graph main_graph for CUDAEExecutionProvider. It might have negative impact on performance (including unable to run CUDA graph). Set session_options.log_severity_level=1 to see the detail logs before this message.
FPS: 44.1233
device=gpu
Model size (on disk): 25.22 MiB (26,448,568 bytes)
mAP@IoU=0.50 : 0.635499

```

640 input size

```

(yolo_quant) PS F:\Final_Lattice_Data_Structure\ActiveMixedMaterialDataset\Suraj_Code_Files\P_Det> python .\code_files\main_execution.py --model "models\yolo_onnx_models\INT8_static_quantized_models\640\yolov8n_640_static_quantized.onnx" --device gpu --input-size 640 --data-root ".\imagedata"
2026-05-04 10:54:09.7148065 [W:onnxruntime:, transformer_memcpy.cc:111 onnxruntime::MemcpyTransformer::ApplyImpl] 63 Mempcy nodes are added to the graph main_graph for CUDAEExecutionProvider. It might have negative impact on performance (including unable to run CUDA graph). Set session_options.log_severity_level=1 to see the detail logs before this message.
FPS: 39.3980
device=gpu
Model size (on disk): 3.43 MiB (3,600,269 bytes)
mAP@IoU=0.50 : 0.658794

(yolo_quant) PS F:\Final_Lattice_Data_Structure\ActiveMixedMaterialDataset\Suraj_Code_Files\P_Det> python .\code_files\main_execution.py --model "models\yolo_onnx_models\INT8_static_quantized_models\640\yolov8s_640_static_quantized.onnx" --device gpu --input-size 640 --data-root ".\imagedata"
2026-05-04 10:54:27.5805485 [W:onnxruntime:, transformer_memcpy.cc:111 onnxruntime::MemcpyTransformer::ApplyImpl] 83 Mempcy nodes are added to the graph main_graph for CUDAEExecutionProvider. It might have negative impact on performance (including unable to run CUDA graph). Set session_options.log_severity_level=1 to see the detail logs before this message.
FPS: 37.0933
device=gpu
Model size (on disk): 11.14 MiB (11,680,726 bytes)
mAP@IoU=0.50 : 0.684049

(yolo_quant) PS F:\Final_Lattice_Data_Structure\ActiveMixedMaterialDataset\Suraj_Code_Files\P_Det> python .\code_files\main_execution.py --model "models\yolo_onnx_models\INT8_static_quantized_models\640\yolov8m_640_static_quantized.onnx" --device gpu --input-size 640 --data-root ".\imagedata"
2026-05-04 10:54:43.5697818 [W:onnxruntime:, transformer_memcpy.cc:111 onnxruntime::MemcpyTransformer::ApplyImpl] 83 Mempcy nodes are added to the graph main_graph for CUDAEExecutionProvider. It might have negative impact on performance (including unable to run CUDA graph). Set session_options.log_severity_level=1 to see the detail logs before this message.
FPS: 30.2650
device=gpu
Model size (on disk): 25.34 MiB (26,574,624 bytes)
mAP@IoU=0.50 : 0.708243

```

1280 input size

```

(yolo_quant) PS F:\Final_Lattice_Data_Structure\ActiveMixedMaterialDataset\Suraj_Code_Files\P_Det> python .\code_files\main_execution.py --model "models\yolo_onnx_models\INT8_static_quantized_models\1280\yolov8n_1280_static_quantized.onnx" --device gpu --input-size 1280 --data-root ".\imagedata"
2026-05-04 10:55:22.0742295 [W:onnxruntime:, transformer_memcpy.cc:111 onnxruntime::MemcpyTransformer::ApplyImpl] 63 Mempcy nodes are added to the graph main_graph for CUDAEExecutionProvider. It might have negative impact on performance (including unable to run CUDA graph). Set session_options.log_severity_level=1 to see the detail logs before this message.
FPS: 16.6463
device=gpu
Model size (on disk): 3.91 MiB (4,104,411 bytes)
mAP@IoU=0.50 : 0.710115

(yolo_quant) PS F:\Final_Lattice_Data_Structure\ActiveMixedMaterialDataset\Suraj_Code_Files\P_Det> python .\code_files\main_execution.py --model "models\yolo_onnx_models\INT8_static_quantized_models\1280\yolov8s_1280_static_quantized.onnx" --device gpu --input-size 1280 --data-root ".\imagedata"
2026-05-04 10:55:46.0572964 [W:onnxruntime:, transformer_memcpy.cc:111 onnxruntime::MemcpyTransformer::ApplyImpl] 63 Mempcy nodes are added to the graph main_graph for CUDAEExecutionProvider. It might have negative impact on performance (including unable to run CUDA graph). Set session_options.log_severity_level=1 to see the detail logs before this message.
FPS: 14.3319
device=gpu
Model size (on disk): 11.62 MiB (12,184,868 bytes)
mAP@IoU=0.50 : 0.733685

(yolo_quant) PS F:\Final_Lattice_Data_Structure\ActiveMixedMaterialDataset\Suraj_Code_Files\P_Det> python .\code_files\main_execution.py --model "models\yolo_onnx_models\INT8_static_quantized_models\1280\yolov8m_1280_static_quantized.onnx" --device gpu --input-size 1280 --data-root ".\imagedata"
2026-05-04 10:56:06.9808738 [W:onnxruntime:, transformer_memcpy.cc:111 onnxruntime::MemcpyTransformer::ApplyImpl] 83 Mempcy nodes are added to the graph main_graph for CUDAEExecutionProvider. It might have negative impact on performance (including unable to run CUDA graph). Set session_options.log_severity_level=1 to see the detail logs before this message.
FPS: 11.0798
device=gpu
Model size (on disk): 25.82 MiB (27,078,806 bytes)
mAP@IoU=0.50 : 0.747494

```

FP32 (CPU2)

320 input size

```
(yolo_quant) PS F:\Final_Lattice_Data_Structure\ActiveMixedMaterialDataset\Suraj_Code_Files\P_Det> python .\code_files\main_execution.py --model "models\yolo_onnx_models\FP32_onnx_models\320\yolov8n_320.onnx" --device cpu --input-size 320 --data-root ".\imagedata"
FPS: 67.3956
device=cpu
Model size (on disk): 12.14 MiB (12,725,005 bytes)
mAP@IoU=0.50 : 0.521438
(yolo_quant) PS F:\Final_Lattice_Data_Structure\ActiveMixedMaterialDataset\Suraj_Code_Files\P_Det> python .\code_files\main_execution.py --model "models\yolo_onnx_models\FP32_onnx_models\320\yolov8s_320.onnx" --device cpu --input-size 320 --data-root ".\imagedata"
FPS: 47.6566
device=cpu
Model size (on disk): 42.67 MiB (44,743,799 bytes)
mAP@IoU=0.50 : 0.614228
(yolo_quant) PS F:\Final_Lattice_Data_Structure\ActiveMixedMaterialDataset\Suraj_Code_Files\P_Det> python .\code_files\main_execution.py --model "models\yolo_onnx_models\FP32_onnx_models\320\yolov8m_320.onnx" --device cpu --input-size 320 --data-root ".\imagedata"
FPS: 29.9098
device=cpu
Model size (on disk): 98.88 MiB (103,683,476 bytes)
mAP@IoU=0.50 : 0.660906
(yolo_quant) PS F:\Final_Lattice_Data_Structure\ActiveMixedMaterialDataset\Suraj_Code_Files\P_Det>
```

640 input size

```
(yolo_quant) PS F:\Final_Lattice_Data_Structure\ActiveMixedMaterialDataset\Suraj_Code_Files\P_Det> python .\code_files\main_execution.py --model "models\yolo_onnx_models\FP32_onnx_models\640\yolov8n_640.onnx" --device cpu --input-size 640 --data-root ".\imagedata"
FPS: 27.9382
device=cpu
Model size (on disk): 12.26 MiB (12,851,087 bytes)
mAP@IoU=0.50 : 0.679877
(yolo_quant) PS F:\Final_Lattice_Data_Structure\ActiveMixedMaterialDataset\Suraj_Code_Files\P_Det> python .\code_files\main_execution.py --model "models\yolo_onnx_models\FP32_onnx_models\640\yolov8s_640.onnx" --device cpu --input-size 640 --data-root ".\imagedata"
FPS: 19.3220
device=cpu
Model size (on disk): 42.79 MiB (44,869,881 bytes)
mAP@IoU=0.50 : 0.707885
(yolo_quant) PS F:\Final_Lattice_Data_Structure\ActiveMixedMaterialDataset\Suraj_Code_Files\P_Det> python .\code_files\main_execution.py --model "models\yolo_onnx_models\FP32_onnx_models\640\yolov8m_640.onnx" --device cpu --input-size 640 --data-root ".\imagedata"
FPS: 10.9177
device=cpu
Model size (on disk): 99.00 MiB (103,809,586 bytes)
mAP@IoU=0.50 : 0.723569
```

1280 input size

```
(yolo_quant) PS F:\Final_Lattice_Data_Structure\ActiveMixedMaterialDataset\Suraj_Code_Files\P_Det> python .\code_files\main_execution.py --model "models\yolo_onnx_models\FP32_onnx_models\1280\yolov8n_1280.onnx" --device cpu --input-size 1280 --data-root ".\imagedata"
FPS: 7.9756
device=cpu
Model size (on disk): 12.74 MiB (13,355,362 bytes)
mAP@IoU=0.50 : 0.725776
(yolo_quant) PS F:\Final_Lattice_Data_Structure\ActiveMixedMaterialDataset\Suraj_Code_Files\P_Det> python .\code_files\main_execution.py --model "models\yolo_onnx_models\FP32_onnx_models\1280\yolov8s_1280.onnx" --device cpu --input-size 1280 --data-root ".\imagedata"
FPS: 5.2124
device=cpu
Model size (on disk): 43.27 MiB (45,374,156 bytes)
mAP@IoU=0.50 : 0.760904
(yolo_quant) PS F:\Final_Lattice_Data_Structure\ActiveMixedMaterialDataset\Suraj_Code_Files\P_Det> python .\code_files\main_execution.py --model "models\yolo_onnx_models\FP32_onnx_models\1280\yolov8m_1280.onnx" --device cpu --input-size 1280 --data-root ".\imagedata"
FPS: 2.8821
device=cpu
Model size (on disk): 99.48 MiB (104,313,941 bytes)
mAP@IoU=0.50 : 0.757325
```

Fp16 quantization (CPU2)

320 input size

```
(yolo_quant) PS F:\Final_Lattice_Data_Structure\ActiveMixedMaterialDataset\Suraj_Code_Files\P_Det> python .\code_files\main_execution.py --model "models\yolo_onnx_models\FP16_quantized_models\320\yolov8n_320_fp16.onnx" --device cpu --input-size 320 --data-root ".\imagedata"
FPS: 57.9423
device=cpu
Model size (on disk): 6.08 MiB (6,373,145 bytes)
mAP@IoU=0.50 : 0.520132
(yolo_quant) PS F:\Final_Lattice_Data_Structure\ActiveMixedMaterialDataset\Suraj_Code_Files\P_Det> python .\code_files\main_execution.py --model "models\yolo_onnx_models\FP16_quantized_models\320\yolov8s_320_fp16.onnx" --device cpu --input-size 320 --data-root ".\imagedata"
FPS: 43.4874
device=cpu
Model size (on disk): 21.35 MiB (22,382,527 bytes)
mAP@IoU=0.50 : 0.614328
(yolo_quant) PS F:\Final_Lattice_Data_Structure\ActiveMixedMaterialDataset\Suraj_Code_Files\P_Det> python .\code_files\main_execution.py --model "models\yolo_onnx_models\FP16_quantized_models\320\yolov8m_320_fp16.onnx" --device cpu --input-size 320 --data-root ".\imagedata"
FPS: 28.3625
device=cpu
Model size (on disk): 49.45 MiB (51,854,796 bytes)
mAP@IoU=0.50 : 0.663455
```

640 input size

```
(yolo_quant) PS F:\Final_Lattice_Data_Structure\ActiveMixedMaterialDataset\Suraj_Code_Files\P_Det> python .\code_files\main_execution.py --model "models\yolo_onnx_models\FP16_quantized_models\640\yolov8n_640_fp16.onnx" --device cpu --input-size 640 --data-root ".\imagedata"
FPS: 22.2073
device=cpu
Model size (on disk): 6.14 MiB (6,436,157 bytes)
mAP@IoU=0.50 : 0.679810
(yolo_quant) PS F:\Final_Lattice_Data_Structure\ActiveMixedMaterialDataset\Suraj_Code_Files\P_Det> python .\code_files\main_execution.py --model "models\yolo_onnx_models\FP16_quantized_models\640\yolov8s_640_fp16.onnx" --device cpu --input-size 640 --data-root ".\imagedata"
FPS: 16.2747
device=cpu
Model size (on disk): 21.41 MiB (22,445,539 bytes)
mAP@IoU=0.50 : 0.707883
(yolo_quant) PS F:\Final_Lattice_Data_Structure\ActiveMixedMaterialDataset\Suraj_Code_Files\P_Det> python .\code_files\main_execution.py --model "models\yolo_onnx_models\FP16_quantized_models\640\yolov8m_640_fp16.onnx" --device cpu --input-size 640 --data-root ".\imagedata"
FPS: 9.5101
device=cpu
Model size (on disk): 49.51 MiB (51,917,808 bytes)
mAP@IoU=0.50 : 0.723518
(yolo_quant) PS F:\Final_Lattice_Data_Structure\ActiveMixedMaterialDataset\Suraj_Code_Files\P_Det>
```

1280 input size

```
(yolo_quant) PS F:\Final_Lattice_Data_Structure\ActiveMixedMaterialDataset\Suraj_Code_Files\P_Det> python .\code_files\main_execution.py --model "models\yolo_onnx_models\FP16_quantized_models\1280\yolov8n_1280_fp16.onnx" --device cpu --input-size 1280 --data-root ".\imagedata"
FPS: 6.3933
device=cpu
Model size (on disk): 6.38 MiB (6,688,163 bytes)
mAP@IoU=0.50 : 0.725676
(yolo_quant) PS F:\Final_Lattice_Data_Structure\ActiveMixedMaterialDataset\Suraj_Code_Files\P_Det> python .\code_files\main_execution.py --model "models\yolo_onnx_models\FP16_quantized_models\1280\yolov8s_1280_fp16.onnx" --device cpu --input-size 1280 --data-root ".\imagedata"
FPS: 4.3144
device=cpu
Model size (on disk): 21.65 MiB (22,697,545 bytes)
mAP@IoU=0.50 : 0.761036
(yolo_quant) PS F:\Final_Lattice_Data_Structure\ActiveMixedMaterialDataset\Suraj_Code_Files\P_Det> python .\code_files\main_execution.py --model "models\yolo_onnx_models\FP16_quantized_models\1280\yolov8m_1280_fp16.onnx" --device cpu --input-size 1280 --data-root ".\imagedata"
FPS: 2.3095
device=cpu
Model size (on disk): 49.75 MiB (52,169,814 bytes)
mAP@IoU=0.50 : 0.758460
(yolo_quant) PS F:\Final_Lattice_Data_Structure\ActiveMixedMaterialDataset\Suraj_Code_Files\P_Det>
```

INT8 dynamic quantization (CPU2)

320 input size

```
(yolo_quant) PS F:\Final_Lattice_Data_Structure\ActiveMixedMaterialDataset\Suraj_Code_Files\P_Det> python .\code_files\main_execution.py --model "models\yolo_onnx_models\INT8_dynamic_quantized_models\320\yolov8n_320_dynamic_quantized.onnx" --device cpu --input-size 320 --data-root ".\imagedata"
FPS: 64.7625
device=cpu
Model size (on disk): 12.14 MiB (12,725,187 bytes)
mAP@IoU=0.50 : 0.521438
(yolo_quant) PS F:\Final_Lattice_Data_Structure\ActiveMixedMaterialDataset\Suraj_Code_Files\P_Det> python .\code_files\main_execution.py --model "models\yolo_onnx_models\INT8_dynamic_quantized_models\320\yolov8s_320_dynamic_quantized.onnx" --device cpu --input-size 320 --data-root ".\imagedata"
FPS: 47.6578
device=cpu
Model size (on disk): 42.67 MiB (44,744,007 bytes)
mAP@IoU=0.50 : 0.614228
(yolo_quant) PS F:\Final_Lattice_Data_Structure\ActiveMixedMaterialDataset\Suraj_Code_Files\P_Det> python .\code_files\main_execution.py --model "models\yolo_onnx_models\INT8_dynamic_quantized_models\320\yolov8m_320_dynamic_quantized.onnx" --device cpu --input-size 320 --data-root ".\imagedata"
FPS: 30.7470
device=cpu
Model size (on disk): 98.88 MiB (103,683,732 bytes)
mAP@IoU=0.50 : 0.660906
(yolo_quant) PS F:\Final_Lattice_Data_Structure\ActiveMixedMaterialDataset\Suraj_Code_Files\P_Det>
```

640 input size

```
(yolo_quant) PS F:\Final_Lattice_Data_Structure\ActiveMixedMaterialDataset\Suraj_Code_Files\P_Det> python .\code_files\main_execution.py --model "models\yolo_onnx_models\INT8_dynamic_quantized_models\640\yolov8n_640_dynamic_quantized.onnx" --device cpu --input-size 640 --data-root ".\imagedata"
FPS: 28.0307
device=cpu
Model size (on disk): 12.26 MiB (12,851,269 bytes)
mAP@IoU=0.50 : 0.679877
(yolo_quant) PS F:\Final_Lattice_Data_Structure\ActiveMixedMaterialDataset\Suraj_Code_Files\P_Det> python .\code_files\main_execution.py --model "models\yolo_onnx_models\INT8_dynamic_quantized_models\640\yolov8s_640_dynamic_quantized.onnx" --device cpu --input-size 640 --data-root ".\imagedata"
FPS: 19.5231
device=cpu
Model size (on disk): 42.79 MiB (44,870,089 bytes)
mAP@IoU=0.50 : 0.707885
(yolo_quant) PS F:\Final_Lattice_Data_Structure\ActiveMixedMaterialDataset\Suraj_Code_Files\P_Det> python .\code_files\main_execution.py --model "models\yolo_onnx_models\INT8_dynamic_quantized_models\640\yolov8m_640_dynamic_quantized.onnx" --device cpu --input-size 640 --data-root ".\imagedata"
FPS: 11.1965
device=cpu
Model size (on disk): 99.00 MiB (103,809,842 bytes)
mAP@IoU=0.50 : 0.723569
(yolo_quant) PS F:\Final_Lattice_Data_Structure\ActiveMixedMaterialDataset\Suraj_Code_Files\P_Det>
```

1280 input size

```
(yolo_quant) PS F:\Final_Lattice_Data_Structure\ActiveMixedMaterialDataset\Suraj_Code_Files\P_Det> python .\code_files\main_execution.py --model "models\yolo_onnx_models\INT8_dynamic_quantized_models\1280\yolov8n_1280_dynamic_quantized.onnx" --device cpu --input-size 1280 --data-root ".\imagedata"
FPS: 7.5783
device=cpu
Model size (on disk): 12.74 MiB (13,355,544 bytes)
mAP@IoU=0.50 : 0.725776
(yolo_quant) PS F:\Final_Lattice_Data_Structure\ActiveMixedMaterialDataset\Suraj_Code_Files\P_Det> python .\code_files\main_execution.py --model "models\yolo_onnx_models\INT8_dynamic_quantized_models\1280\yolov8s_1280_dynamic_quantized.onnx" --device cpu --input-size 1280 --data-root ".\imagedata"
FPS: 5.0016
device=cpu
Model size (on disk): 43.27 MiB (45,374,364 bytes)
mAP@IoU=0.50 : 0.760904
(yolo_quant) PS F:\Final_Lattice_Data_Structure\ActiveMixedMaterialDataset\Suraj_Code_Files\P_Det> python .\code_files\main_execution.py --model "models\yolo_onnx_models\INT8_dynamic_quantized_models\1280\yolov8m_1280_dynamic_quantized.onnx" --device cpu --input-size 1280 --data-root ".\imagedata"
FPS: 2.5723
device=cpu
Model size (on disk): 99.48 MiB (104,314,197 bytes)
mAP@IoU=0.50 : 0.757325
(yolo_quant) PS F:\Final_Lattice_Data_Structure\ActiveMixedMaterialDataset\Suraj_Code_Files\P_Det>
```

INT8 Static quantization (CPU2)

320 input size

```
(yolo_quant) PS F:\Final_Lattice_Data_Structure\ActiveMixedMaterialDataset\Suraj_Code_Files\P_Det> python .\code_files\main_execution.py --model "models\yolo_onnx_models\INT8_static_quantized_models\320\yolov8n_320_static_quantized.onnx" --device cpu --input-size 320 --data-root ".\imagedata"
FPS: 48.3574
device=cpu
Model size (on disk): 3.31 MiB (3,474,227 bytes)
mAP@IoU=0.50 : 0.497910
(yolo_quant) PS F:\Final_Lattice_Data_Structure\ActiveMixedMaterialDataset\Suraj_Code_Files\P_Det> python .\code_files\main_execution.py --model "models\yolo_onnx_models\INT8_static_quantized_models\320\yolov8s_320_static_quantized.onnx" --device cpu --input-size 320 --data-root ".\imagedata"
FPS: 29.5353
device=cpu
Model size (on disk): 11.02 MiB (11,554,684 bytes)
mAP@IoU=0.50 : 0.564358
(yolo_quant) PS F:\Final_Lattice_Data_Structure\ActiveMixedMaterialDataset\Suraj_Code_Files\P_Det> python .\code_files\main_execution.py --model "models\yolo_onnx_models\INT8_static_quantized_models\320\yolov8m_320_static_quantized.onnx" --device cpu --input-size 320 --data-root ".\imagedata"
FPS: 16.4254
device=cpu
Model size (on disk): 25.22 MiB (26,448,568 bytes)
mAP@IoU=0.50 : 0.628468
(yolo_quant) PS F:\Final_Lattice_Data_Structure\ActiveMixedMaterialDataset\Suraj_Code_Files\P_Det>
```

640 input size

```
(yolo_quant) PS F:\Final_Lattice_Data_Structure\ActiveMixedMaterialDataset\Suraj_Code_Files\P_Det> python .\code_files\main_execution.py --model "models\yolo_onnx_models\INT8_static_quantized_models\640\yolov8n_640_static_quantized.onnx" --device cpu --input-size 640 --data-root ".\imagedata"
FPS: 18.7071
device=cpu
Model size (on disk): 3.43 MiB (3,600,269 bytes)
mAP@IoU=0.50 : 0.651404
(yolo_quant) PS F:\Final_Lattice_Data_Structure\ActiveMixedMaterialDataset\Suraj_Code_Files\P_Det> python .\code_files\main_execution.py --model "models\yolo_onnx_models\INT8_static_quantized_models\640\yolov8s_640_static_quantized.onnx" --device cpu --input-size 640 --data-root ".\imagedata"
FPS: 11.5520
device=cpu
Model size (on disk): 11.14 MiB (11,680,726 bytes)
mAP@IoU=0.50 : 0.601228
(yolo_quant) PS F:\Final_Lattice_Data_Structure\ActiveMixedMaterialDataset\Suraj_Code_Files\P_Det> python .\code_files\main_execution.py --model "models\yolo_onnx_models\INT8_static_quantized_models\640\yolov8m_640_static_quantized.onnx" --device cpu --input-size 640 --data-root ".\imagedata"
FPS: 6.2983
device=cpu
Model size (on disk): 25.34 MiB (26,574,624 bytes)
mAP@IoU=0.50 : 0.715222
(yolo_quant) PS F:\Final_Lattice_Data_Structure\ActiveMixedMaterialDataset\Suraj_Code_Files\P_Det>
```

1280 input size

```
(yolo_quant) PS F:\Final_Lattice_Data_Structure\ActiveMixedMaterialDataset\Suraj_Code_Files\P_Det> python .\code_files\main_execution.py --model "models\yolo_onnx_models\INT8_static_quantized_models\1280\yolov8n_1280_static_quantized.onnx" --device cpu --input-size 1280 --data-root ".\imagedata"
FPS: 6.0194
device=cpu
Model size (on disk): 3.91 MiB (4,104,411 bytes)
mAP@IoU=0.50 : 0.704724
(yolo_quant) PS F:\Final_Lattice_Data_Structure\ActiveMixedMaterialDataset\Suraj_Code_Files\P_Det> python .\code_files\main_execution.py --model "models\yolo_onnx_models\INT8_static_quantized_models\1280\yolov8s_1280_static_quantized.onnx" --device cpu --input-size 1280 --data-root ".\imagedata"
FPS: 3.3785
device=cpu
Model size (on disk): 11.62 MiB (12,184,868 bytes)
mAP@IoU=0.50 : 0.734559
(yolo_quant) PS F:\Final_Lattice_Data_Structure\ActiveMixedMaterialDataset\Suraj_Code_Files\P_Det> python .\code_files\main_execution.py --model "models\yolo_onnx_models\INT8_static_quantized_models\1280\yolov8m_1280_static_quantized.onnx" --device cpu --input-size 1280 --data-root ".\imagedata"
FPS: 1.7779
device=cpu
Model size (on disk): 25.82 MiB (27,078,806 bytes)
mAP@IoU=0.50 : 0.757150
```